

Expected-Return Policy Inference with Guided Replica Exchange

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Abstract

We study generalized Bayesian inference over complete deterministic policies while preserving the ordinary expected-return objective of a stochastic Markov decision process. The target is a Gibbs measure whose log density is proportional to a policy’s expected return. It is therefore different from a trajectory target that averages exponentiated realized returns, which is risk-sensitive even when the intended control objective is not. Our primary algorithm is replica-exchange Metropolis–Hastings in policy space. A learned PPO policy supplies full-support single-state proposals, but an exact Hastings correction prevents the guide from changing the invariant distribution. When the transition model is available, dynamic programming evaluates the exact target; with simulator-only access, an immutable bank of common random-number tapes defines a transparent sample-average approximation. Direct variational and replicated sequential Monte Carlo estimators provide complementary ablations. Evaluation samples one complete policy and commits to it for the episode, and compares against precisely specified REINFORCE, PPO-Clip, discrete SAC, categorical cross-entropy search, and Double Q-learning baselines under matched simulator access and compute accounting in a separate information stratum. Exact finite-policy gates precede larger stochastic GridWorld experiments. Confirmatory empirical results are not yet claimed in this draft; medium-scale mixing and cost gates remain in progress.

1 Introduction

Planning in a stochastic environment requires two conceptually separate operations. First, a policy is evaluated by averaging its return over the environment’s transition randomness. Second, a decision rule selects among policies. Conflating these operations changes the objective: exponentiating a single realized return before averaging is generally not equivalent to exponentiating the expected return.

This manuscript develops a policy-inference formulation in which the target is defined before an approximation algorithm is chosen. The target assigns mass to complete deterministic policies according to expected return. Its modes are therefore expected-return maximizers, while a finite inverse temperature retains a distribution over alternatives. The construction is a generalized Gibbs posterior over decisions, not a posterior belief about unknown transition parameters. Related free-utility formulations also trade expected utility against information cost (Ortega and Braun, 2011), and recent general Bayesian policy learning updates distributions over decision rules from a loss (Kato, 2026). Our setting differs by targeting finite-horizon MDP policies and by making the expected-return order of operations explicit.

Control-as-inference methods provide a broad connection between optimal control and probabilistic inference (Levine, 2018). In stochastic dynamics, however, the choice of graphical model and optimality likelihood determines whether the resulting objective is risk-neutral, risk-sensitive, or maximum-entropy. We therefore do not treat “inference” as sufficient to identify the control objective.

The study makes five contributions:

1. It specifies an expected-return Gibbs measure over complete policies, including its reference measure, temperature, and execution semantics.
2. It gives exact finite-policy oracles and variance-sensitive diagnostics that detect substitution of an exponentiated-trajectory objective.
3. It introduces guided replica-exchange policy MCMC whose PPO guide is fully corrected by Metropolis–Hastings and therefore affects mixing, not the invariant target.
4. It separates exact model evaluation from a fixed-tape simulator approximation, and retains direct variational and replicated SMC alternatives as controlled ablations.
5. It defines a fail-fast comparison against five published algorithmic families with common policy-commitment semantics and full cost accounting.

Agreement with exact enumeration establishes target correctness before any large-scale performance comparison is attempted. The experimental protocol is specified before confirmatory results; this draft labels incomplete empirical claims as preliminary or to be determined.

2 Problem Setting and Notation

2.1 Finite-horizon stochastic control

Let an episodic Markov decision process be $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \rho_0, H, \gamma)$, where $T(\cdot \mid s, a)$ is a transition kernel, $R(s, a, s')$ is a bounded reward, ρ_0 is the initial-state distribution, and H is the rollout horizon. The discount is $\gamma \in (0, 1]$; the current GridWorld study uses $\gamma = 1$.

For a deterministic policy $\pi \in \Pi$, a trajectory $\tau = (s_1, a_1, \dots, s_{H+1})$ is generated by

$$s_1 \sim \rho_0, \quad a_t = \pi(s_t), \quad s_{t+1} \sim T(\cdot \mid s_t, a_t).$$

Its truncated return and the policy value are

$$G_H(\tau) = \sum_{t=1}^H \gamma^{t-1} R(s_t, a_t, s_{t+1}), \quad J_H(\pi) = \mathbb{E}_{\tau \sim p(\cdot \mid \pi)}[G_H(\tau)]. \quad (1)$$

The implemented policy class is stationary in the physical state. Time-to-go is deliberately not appended to the observation, so recurrent states at different remaining horizons can be aliased. Every learned and inferred method shares this restriction. Finite-horizon dynamic programming is reported as a nonstationary upper control, not as a like-for-like competitor. A future time-indexed study must augment the state for every method rather than only for the proposed approach.

2.2 Policy-level randomness

A distribution over deterministic policies is different from a Markov stochastic policy. Sampling one deterministic policy at the start of an episode creates correlations across states and revisits. Sampling a new action from a statewise marginal on every visit removes those correlations. Both execution rules can be studied, but they must not be treated as interchangeable. Our primary rule is *sample then commit*: draw all state actions defining one deterministic policy before evaluation and preserve them throughout the episode. Per-visit marginal execution and the statewise maximum-probability policy are secondary diagnostics.

Assumption 2.1 (Reference measure). *The deterministic policy class has a specified probability measure μ . For a finite tabular class, μ may be uniform. For a structured or continuous class, the choice of μ is part of the model and must be reported.*

3 Expected-Return Policy Target

For inverse temperature $\beta > 0$, define the target measure

$$P_\beta(d\pi) = \frac{\exp(\beta J_H(\pi))}{Z_\beta} \mu(d\pi), \quad Z_\beta = \int_{\Pi} \exp(\beta J_H(\pi)) \mu(d\pi). \quad (2)$$

Bounded rewards and a finite reference measure make Z_β finite. The parameter β carries the inverse units of reward, so a change of reward units must be accompanied by the inverse change in β . The pair (μ, β) is part of the problem specification: neither is identified by simulator data alone. For the tabular experiments, μ is uniform over complete stationary deterministic policies.

Proposition 3.1 (Mode preservation). *Every mode of P_β with respect to a uniform finite reference measure maximizes J_H , and every expected-return maximizer is a mode.*

Proof. The exponential function is strictly increasing, and the normalizing constant does not depend on π . $\square$

The order of expectation and exponentiation is essential. In general,

$$\exp(\beta \mathbb{E}[G_H]) \neq \mathbb{E}[\exp(\beta G_H)]. \quad (3)$$

The right-hand side defines an entropic-risk quantity and places additional weight on return variability. By Jensen’s inequality it is at least the left-hand side for $\beta > 0$, with strict inequality for nonconstant return. It is not an estimator identity for the target in Equation (2).

The target also has the variational characterization

$$\log Z_\beta = \sup_Q \{ \beta \mathbb{E}_{\pi \sim Q}[J_H(\pi)] - \text{KL}(Q \parallel \mu) \}. \quad (4)$$

Thus the regularization acts on a distribution over complete policies. It is not the same as adding independent action entropy at every visited state.

Equation (2) may be read as a generalized posterior whose loss is $-J_H$. This interpretation is useful algorithmically, but it does not imply epistemic uncertainty calibration or Bayesian learning of the MDP. At $\beta \rightarrow \infty$, mass concentrates on expected-return maximizers; at finite β , output quality and diversity depend on both the temperature and the policy reference measure.

4 Exact Validation Framework

Correct implementation of an MCMC transition does not establish that the intended density was specified. Validation therefore starts with finite MDPs for which all $|\mathcal{A}|^D$ stationary deterministic policies can be enumerated, where D is the number of decision states. Finite-horizon dynamic programming computes $J_H(\pi)$ for every policy, and log-sum-exp normalization gives the exact target probabilities, partition function, and action marginals in Equation (2).

4.1 Target-substitution diagnostic

The smallest diagnostic contains two policies with equal expected return but different return variance. Under a uniform reference measure, Equation (2) assigns them equal probability. A method that averages $\exp(\beta G_H)$ instead assigns more mass to the variable policy. This is a deterministic pass-or-fail test of the order of operations rather than a performance benchmark.

Additional exact tests cover state revisitation, absorbing states, finite horizons, policy memoization, common random numbers, Metropolis–Hastings proposal ratios, replica swaps,

particle ancestry, and score-function gradients. In particular, empirical frequencies of both single-chain and replica-exchange samplers must agree with the enumerated target on tiny MDPs.

4.2 Fail-fast gates

Experiments advance only after the preceding gate passes:

- G1. **Mechanics.** Transition rows, rewards, termination, horizons, seeded replay, artifact writing, and budget counters pass unit tests.
- G2. **Target.** Enumeration verifies exact probabilities, modes, marginals, local acceptance ratios, and swap acceptance ratios.
- G3. **Approximation.** Increasing the immutable tape-bank size reduces sample-average target error on repeated small problems; a failure stops simulator-only scaling.
- G4. **Mixing.** A one-chain rescue check first requires an accepted high-temperature whole-policy refresh, an accepted final-edge swap, and more than one cold-reaching walker. A cheap two-chain ladder-flow probe then requires post-burn acceptance of at least 0.20 on every edge and an endpoint transition in each chain. Passing this probe authorizes, but does not replace, the four-chain mixing run. That full gate requires split return $\hat{R} \leq 1.05$, per-chain return ESS of at least 100, flow through every ladder edge and in both directions, between-chain coordinate agreement, and within-chain policy movement. High local acceptance or either passing fail-fast probe alone is not evidence of mixing.
- G5. **Cost.** Three short runs establish memory, wall-clock time, and value-evaluation or simulator-step rates before a large job array is submitted.

The final thresholds for effective sample size and target error are frozen after pilot calibration and before confirmatory seeds are inspected. Failed, timed-out, and numerically invalid runs remain in the accounting.

5 Expected-Return Policy Inference

5.1 Exact and sample-average policy scores

The sampler uses a score $\tilde{J}(\pi)$. With an explicit finite transition model, $\tilde{J} = J_H$ is evaluated exactly by dynamic programming. This variant targets Equation (2) exactly, but is model-based.

With simulator-only access, draw an immutable bank of K random-number tapes $\Xi_K = (\xi_1, \dots, \xi_K)$ and define

$$\hat{J}_K(\pi; \Xi_K) = \frac{1}{K} \sum_{k=1}^K G_H(\pi, \xi_k). \quad (5)$$

Conditional on that bank, the implemented target is

$$P_{\beta, K}(d\pi \mid \Xi_K) \propto \exp\left(\beta \hat{J}_K(\pi; \Xi_K)\right) \mu(d\pi). \quad (6)$$

The same tape bank is reused for every policy evaluation in a chain. This is both common-random-number variance reduction and a definition of the finite-sample target. Refreshing the bank inside an ordinary Metropolis–Hastings ratio would generally invalidate detailed balance. For finite policy spaces and bounded returns, $\hat{J}_K$ converges uniformly to J , so Equation (6) converges to Equation (2); finite K is nevertheless an approximation, not an unbiased exact-target construction.

5.2 A learned proposal that cannot change the target

PPO-Clip (Schulman et al., 2017) is trained first and then frozen. Its statewise action probabilities guide local moves, but PPO is neither the target nor an uncorrected component of the final sampler. Let $p_\phi(a | s)$ be the frozen probabilities and define

$$g_{\eta,\phi}(a | s) = \eta p_\phi(a | s) + (1 - \eta) \frac{1}{|\mathcal{A}|}, \quad 0 \leq \eta < 1. \quad (7)$$

The uniform component gives every action positive probability. If the current action is a , a proposed different action a' has probability

$$q_s(a' | a) = \frac{g_{\eta,\phi}(a' | s)}{1 - g_{\eta,\phi}(a | s)}, \quad a' \neq a. \quad (8)$$

Both the guide snapshot and η are stored with the run. Setting $\eta = 0$ gives the uniform-proposal ablation.

5.3 Single-site policy Metropolis–Hastings

One local proposal selects a decision state uniformly, changes only its action according to Equation (8), and leaves all other policy entries fixed. At inverse temperature b , accept π' from π with probability

$$\alpha_b(\pi, \pi') = \min \left\{ 1, \exp \left[b \{ \tilde{J}(\pi') - \tilde{J}(\pi) \} \right] \frac{\mu(\pi')}{\mu(\pi)} \frac{q_s(a | a')}{q_s(a' | a)} \right\}. \quad (9)$$

The state-selection terms cancel. The experiments use uniform μ , so its ratio is one. The remaining Hastings ratio is essential because forcing a different action makes even a fixed categorical guide asymmetric. With full support, this kernel is irreducible on the finite policy space and satisfies detailed balance with the chosen exact or fixed-tape target.

5.4 Exact whole-policy independence refresh

Single-site edits can be ineffective on a sparse-return plateau, where many policies have the same score but reaching a useful policy requires several coordinated action changes. Each active replica update therefore chooses the single-site kernel with probability $1 - \rho$, or an independence proposal with probability ρ . The fixed whole-policy proposal is

$$Q_\phi(\pi) = w_{\text{map}} \mathbf{1}\{\pi = \pi_\phi^{\text{MAP}}\} + w_{\text{guide}} \prod_{s \in \mathcal{D}} p_\phi(\pi(s) | s) + w_{\text{uniform}} |\mathcal{A}|^{-|\mathcal{D}|}, \quad (10)$$

where $\mathcal{D}$ is the decision-state set and π_ϕ^{MAP} is the statewise mode of the frozen PPO guide. The defaults are $\rho = 0.10$ and $(w_{\text{map}}, w_{\text{guide}}, w_{\text{uniform}}) = (0.25, 0.70, 0.05)$. The components overlap, so the proposal density is the sum in Equation (10), evaluated by log-sum-exp, not merely the density of the component that generated the draw. The positive uniform component guarantees full support.

At inverse temperature b , an independent draw $\pi' \sim Q_\phi$ is accepted with

$$\begin{aligned} \log R_b(\pi, \pi') &= b \{ \tilde{J}(\pi') - \tilde{J}(\pi) \} \\ &\quad + \log \mu(\pi') - \log \mu(\pi) \\ &\quad + \log Q_\phi(\pi) - \log Q_\phi(\pi'), \end{aligned} \quad (11)$$

$$\alpha_b^{\text{global}}(\pi, \pi') = \min \{ 1, \exp[\log R_b(\pi, \pi')] \}. \quad (12)$$

This is an ordinary independence Metropolis–Hastings step (Mengersen and Tweedie, 1996). The complete proposal ratio is mandatory. Consequently, the guide, its mode, and the branch probability affect efficiency but cannot change the invariant target. Full support gives an asymptotic finite-space convergence guarantee, not evidence that a finite run has mixed.

Forced-different proposals can be periodic in a two-action domain. We therefore mix every complete within-replica-plus-swap sweep with the identity kernel:

$$K_\delta = \delta I + (1 - \delta)K, \quad 0 < \delta < 1. \quad (13)$$

The stay-put decision is independent of the chain state, so it preserves the product target while giving every augmented chain state a positive self-transition. The default is $\delta = 0.05$; swap parity advances only on active sweeps. The branch settings, local and global proposal/acceptance counts, δ , and the realized lazy-sweep count are stored with each run.

5.5 Replica exchange across inverse temperatures

Single-site moves can mix slowly when good complete policies occupy separated modes. Replica exchange, also called parallel tempering, couples chains at inverse temperatures

$$0 = b_0 < b_1 < \cdots < b_{M-1} = \beta, \quad b_\ell^{(0)} = \beta\{\ell/(M-1)\}^p, \quad (14)$$

where $p > 0$ controls the initial ladder spacing. Development-only pilots may replace this power ladder with a fixed, explicitly listed ladder that places more replicas at measured exchange bottlenecks, following the feedback principle of Katzgraber et al. (2006). Adaptation stops before target samples are collected, and the endpoints remain zero and β . Replica ℓ applies the fixed mixture of Equations (9) and (12) with $b = b_\ell$. At a declared interval, four checkerboard swap sub-sweeps are applied by default, with parity alternating globally. These extra swaps reuse cached values and compose invariant kernels, so they can improve temperature-space flow without extra policy evaluations (Chodera and Shirts, 2011). For replicas $i < j$, the swap acceptance probability is

$$\alpha_{i,j}^{\text{swap}} = \min\left\{1, \exp\left[(b_j - b_i)\{\tilde{J}(\pi_i) - \tilde{J}(\pi_j)\}\right]\right\}. \quad (15)$$

This is a Metropolis update for the product of tempered targets

$$\prod_{\ell=0}^{M-1} \mu(\pi_\ell) \exp(b_\ell \tilde{J}(\pi_\ell)),$$

and hence preserves the desired cold marginal at $b_{M-1} = \beta$ (Geyer, 1991). The PPO guide affects only proposal efficiency; Equations (9), (12), and (15) retain the target.

By default, all replicas start at the PPO statewise mode, while the $b_0 = 0$ chain is allowed to forget that initializer during burn-in. A separate overdispersed diagnostic initializes hotter replicas from the policy prior. Initialization cannot alter the stationary distribution, but it can strongly affect a finite run. Only thinned post-burn-in draws from the b_{M-1} replica are reported as target samples.

5.6 Diagnostics and cost accounting

Each run records the beta ladder, local and global proposals and acceptances, adjacent swap proposals and acceptances for both the complete and post-burn windows, retained returns, return effective sample size, policy-value evaluations, lazy sweeps, simulator transitions, PPO training transitions, wall-clock time, and software and source hashes. Return effective sample size is necessary but can miss movement between policies with similar value. We therefore

also record action-indicator effective sample sizes, statewise switch rates, distinct sampled policies, walker temperature visits, endpoint transitions, round trips, and the number of walkers reaching the cold replica. Independent warm and overdispersed chains provide the between-chain check. Calibration uses two distinct decisions. A short two-chain probe tests only whether the proposed ladder has viable post-burn edge exchange and endpoint flow; passing it merely permits the longer run. The full multichain gate then tests return $\hat{R}$ and ESS, edge exchange, cold-reaching walkers and round trips, coordinate total variation, unique-policy fraction, mean policy Hamming movement, and switching at nondegenerate coordinates. Only the latter supports a mixing claim. A high-return cold replica cannot compensate for a failure at either stage.

For I total iterations, N_{lazy} lazy sweeps, and M temperatures, the sampler uses $M + M(I - N_{\text{lazy}})$ policy-value evaluations. Swaps reuse cached values.

PPO training is part of the proposed method’s cost. Matched-budget comparisons therefore include every interaction and optimizer update used to train the guide plus all downstream policy evaluations. Exact dynamic-programming evaluations are logged separately because they consume model computations but zero simulator transitions.

5.7 Complementary estimators and ablations

Two additional estimators test which part of the construction matters. First, for a tractable distribution q_θ over complete policies, the direct variational objective is

$$\mathcal{L}(\theta) = \beta \mathbb{E}_{\pi \sim q_\theta} [J_H(\pi)] - \text{KL}(q_\theta \| \mu). \quad (16)$$

Sampling $\pi \sim q_\theta$ and then a rollout conditional on that same policy gives an unbiased estimator of the return term because it is linear in J . A score-function contribution is

$$(\beta G_H + \log \mu(\pi) - \log q_\theta(\pi) - c) \nabla_\theta \log q_\theta(\pi), \quad (17)$$

where c is an action-independent control variate.

Second, replicated policy SMC instantiates each state action once in a lazy policy memo and advances K simulator replicas under that same policy. For newly exposed states U_t , its incremental log weight is

$$\Delta \log w_t = \frac{\beta}{K} \sum_{k=1}^K r_{t,k} + \sum_{s \in U_t} [\log \mu(a_s | s, m_{<s}) - \log q(a_s | s, m_{<s})]. \quad (18)$$

The mean reward is formed before exponentiation, proposal correction is applied once per newly assigned policy entry, revisits reuse the memo, and resampling copies the entire ancestry-dependent state. Sequential Monte Carlo can define powerful variational families (Naesseth et al., 2018) and particle objectives (Maddison et al., 2017), but those results do not by themselves make an exponentiated noisy return an unbiased estimator of Equation (2). Here replicated SMC is evaluated against the explicitly stated fixed-tape target and against exact enumeration.

The isolation matrix contains uniform-guide MCMC, corrected PPO-guided MCMC, PPO alone, direct ELBO, uniform-proposal replicated SMC, and corrected guided SMC. No condition may silently reuse extra simulator interactions.

6 Experimental Design

The empirical study asks four separate questions: (i) does an implementation sample the stated target, (ii) does tempering mix across relevant policy modes, (iii) do target samples

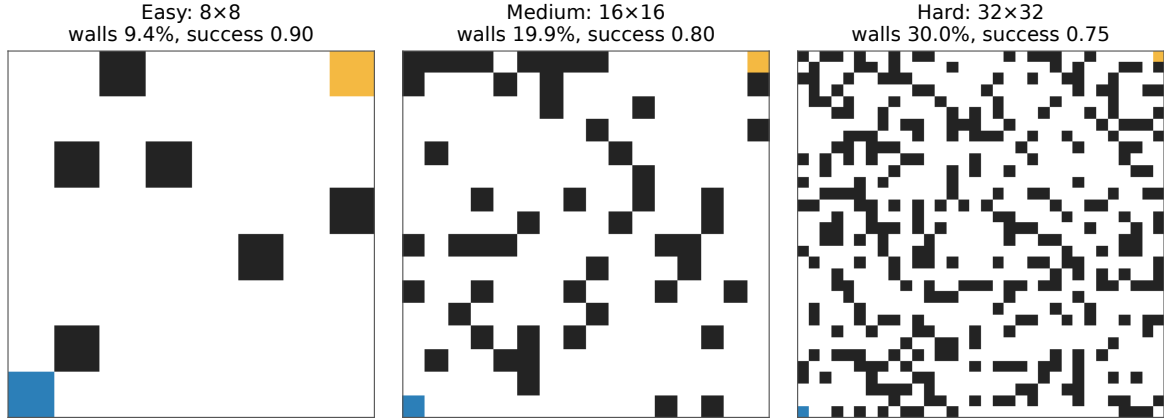


Figure 1: Representative development maps for the three GridWorld tiers. White cells are traversable, dark cells are walls, blue marks the start, and gold marks the goal. Held-out maps are frozen separately and remain unused during method development and tuning.

Table 1: Domain-extension decision. Extensions are conditional on the primary GridWorld study rather than required to enlarge the result table.

Domain	Added evidence	Decision
Blackjack	Compact nonspatial stochasticity and intrinsic return variance with two actions (Towers et al., 2025)	First low-cost extension after GridWorld
Tireworld	Stochastic failures, resources, sparse success, and competing routes (Bonet and Geffner, 2005)	Best second headline domain if time and compute remain
Taxi	Action masks and subgoals, but mostly deterministic and spatial	Optional smoke test only
Academic Advising	Factored multivalued state and concurrent actions (Guerin et al., 2012)	Future structured-policy work
Continuous control	Requires a new reference measure and policy-space sampler	Out of scope

control the environment well under policy commitment, and (iv) what simulator and compute cost is required? Success on the control metric cannot substitute for failure on target fidelity or mixing.

6.1 Domains and scope

Stochastic GridWorld is the complete primary study. Easy, medium, and hard families use, respectively, 8×8 , 16×16 , and 32×32 grids; wall fractions of 9.4%, 19.9%, and 30.0%; action success probabilities of 0.90, 0.80, and 0.75; and horizons 64, 160, and 384. Frozen-map audits record reachability, shortest-path length and count, and single-cell start-goal bottlenecks. Seed zero maps are development-only; held-out maps are frozen before confirmation. Small instances retain exact enumeration; larger instances retain exact dynamic-programming evaluation of any fixed policy even when enumerating all policies is impossible.

Only domains that add a distinct scientific stress test are considered after GridWorld passes its medium gates:

A Tireworld extension would use a verified factored-state adapter; pyRDDLgym is a possible RDDL-to-simulator route (Taitler et al., 2022). Transition, termination, reward, and action-mask tests on a small instance are mandatory before any such results are included.

Table 2: Learned competitors. Uniform random behavior and exact finite-horizon dynamic programming are controls, not competitors.

Reported method	Implemented variant	Role
REINFORCE	Monte Carlo policy gradient with learned state-value baseline (Williams, 1992)	Basic on-policy baseline
PPO-Clip	Clipped surrogate, Monte Carlo advantages, entropy bonus; no GAE or value clipping (Schulman et al., 2017)	Stronger on-policy baseline and separately costed proposal trainer
Discrete SAC	Categorical actor, twin critics, target critics, replay, and fixed entropy temperature (Christodoulou, 2019)	Off-policy entropy-regularized baseline
Categorical CEM	Distribution search over complete stationary deterministic policies (Mannor et al., 2003)	Direct black-box policy search
Double Q-learning	Online tabular Double Q-learning with linearly annealed epsilon-greedy exploration (van Hasselt, 2010)	Value-learning baseline

6.2 Competitors and controls

The competitor implementations preserve each published core update while using a common tabular code path for identical environment semantics, seeding, transition accounting, and artifacts. They are named by the exact implemented variant rather than by a broader software package.

The exact control is nonstationary and may use the known model, so it is an upper reference rather than a fair learned competitor. All learned and inferred output policies use stationary physical-state inputs. For correct finite-horizon backup targets, discrete SAC uses time-indexed twin critics and replay tuples, while Double Q-learning uses time-indexed tables and projects its result into the declared stationary policy class using state-time visitation.

Comparisons are stratified by information access. In the known-model stratum, exact-value tempering is tested for target fidelity and mixing against exact enumeration or known-model controls. In the simulator-only stratum, fixed-tape sample-average tempering and the five learned competitors receive the same generative simulator and matched training-transition allowance. Exact-model tempering may be displayed beside simulator-only learners as a descriptive reference, but such a cross-stratum table cannot support a sample-efficiency superiority claim. Model value evaluations, simulator transitions, and guide training transitions are therefore reported in separate cost columns.

6.3 Common evaluation semantics

For every non-oracle method, the primary evaluator samples a complete deterministic policy from the reported distribution and commits to it for the entire episode. For an explicit particle or MCMC sample, that complete policy is used directly. For statewise categorical outputs, one action is sampled once at every decision state before evaluation. Primary metrics are mean exact expected return and exact goal-reaching probability over these committed policies. Their sample standard deviation and standard error describe variation over policy draws.

Per-visit stochastic execution, statewise marginal probabilities, and the statewise maximum-probability policy are reported separately. They cannot replace the committed-policy result because they destroy or collapse policy-level dependence. Secondary metrics include lower-tail return, distributional target error where enumeration is possible, and mode recovery.

6.4 Fair budgets and tuning

Every comparison uses the same maps, training seeds, evaluation streams, horizon, reward, and termination convention. Current GridWorld runs use $\gamma = 1$; discounted baseline semantics must be validated before any $\gamma < 1$ experiment. Method-specific tuning grids are allowed because a single global learning rate is not a fair tuning policy, but tuning allowance is matched in environment transitions and core-hours. Tuning maps and seeds are disjoint from the final confirmatory set.

The inverse temperature defines the target and is not selected after observing held-out control performance. Exact diagnostic maps use a predeclared sensitivity grid; the headline GridWorld target freezes one value on seed-zero development maps. The current development value is $\beta = 64$. Any additional value is reported as target sensitivity with separate mixing and cost diagnostics, not as a best-of-grid result.

All interactions used by PPO guide training, SMC, policy-bank evaluation, and sample-average MCMC count toward the proposed method. Exact policy-value evaluations, optimizer updates, wall time, CPU or accelerator time, and peak memory are reported separately. A generally applicable exploration aid is offered to compatible methods or labeled as a separate assisted ablation. Competitor bugs are fixed; no baseline is weakened to improve a comparison.

6.5 Cost-effective execution order

After gates G0–G3 in Section 4, one seed and the smallest budget run for every method on each map family. Only finite, semantically valid runs advance to a three-seed timing and learning pilot. That pilot estimates resource requests and identifies clearly dominated settings. Hyperparameters are then tuned on designated development instances. Finally, one frozen manifest launches the full seed-by-map comparison. Job arrays are capped to avoid simultaneous infrastructure failures, and each completed run writes a validated marker so interrupted arrays can resume without overwriting results.

The same order applies to domain extension: one exact Blackjack instance, one seed per method, then three-seed timing, and only then a complete table. Tireworld work begins only after the GridWorld confirmatory manifest is frozen.

6.6 Current empirical status

Confirmatory values are **TBD**. Preliminary medium-GridWorld exact-value tempering diagnostics produced strong committed policies but exposed a sharp intermediate-temperature exchange bottleneck. Four independent 32-temperature, power-four chains, including an overdispersed chain, each retained 800 draws. They failed the predeclared gate: split return $\hat{R} = 1.096$, minimum per-chain return ESS of 10.5, minimum whole-run adjacent-swap acceptance of 0.049, no completed round trip, and maximum pairwise state-action total variation of 0.699. The principal warm-chain bottleneck lay between inverse temperatures 1.44 and 1.98. Because the legacy acceptance counter included burn-in, subsequent runs also store a separate post-burn rate; the ESS, round-trip, and between-chain failures do not depend on that counter. A subsequent two-chain, 4,000-iteration custom-ladder probe also failed. Its warm chain had post-burn acceptance between 0.598 and 0.997 on every edge, but no walker traversed both endpoints. In the overdispersed chain, replicas below the cold endpoint remained on a flat random-policy plateau near return -16 , while the cold replica remained near return 1.1 ; the final edge consequently accepted no swap. High acceptance inside the bad plateau therefore did not indicate useful mixing.

The next predeclared checkpoint tests the exact whole-policy refresh in Equation (12), keeping the same ladder. One overdispersed 1,000-iteration run must accept a post-burn global proposal in the upper ladder quarter, accept a post-burn swap on the final edge, and

Table 3: Required confirmatory reporting. No entry is populated from pilot runs.

Question	Comparison	Primary report
Target correctness	exact enumeration	distribution error and mode mass
Mixing	chains and ladders	ESS, swaps, and between-chain agreement
Control quality	five competitors	committed return and goal probability
Robustness	map families and seeds	uncertainty and failed-run count
Scalability	matched accounting	interactions, time, and memory

send at least two walkers to the cold replica. Only then are one warm and one overdispersed 2,500-iteration chain run; both must traverse the ladder and every post-burn edge must accept at least 0.20. These are tuning diagnostics, not evidence that the method outperforms a competitor. They do not relax the later $\hat{R}$, ESS, round-trip, coordinate-agreement, and within-chain movement gate. Full-scale comparison arrays remain paused until that full gate passes.

7 Discussion and Limitations

The target in Equation (2) has a useful property: with a uniform reference measure, its modes preserve ordinary expected-return optimality. A finite-temperature draw is nevertheless not a classical optimal policy. It is a randomized selection rule controlled by β and μ . The resulting spread should not be described as calibrated epistemic uncertainty about an unknown environment.

Replica exchange corrects neither a misspecified target nor poor finite-run mixing. Local state changes can leave high-dimensional policy modes separated by low-value barriers, a PPO guide can be overconfident in one region, and a temperature ladder can have a bottleneck near the cold chain. Exact Hastings correction guarantees invariance, not useful effective sample size. Swap rates on every edge, multiple dispersed chains, policy-coordinate diagnostics, and explicit nonconvergence decisions are therefore part of the method rather than optional presentation details.

Sparse rewards are a general exploration difficulty (Andrychowicz et al., 2017). Here they also create broad policy plateaus: when two policies have the same score, changing the inverse temperature cannot prefer one over the other. The whole-policy refresh can cross such a plateau without changing the target, but an independence sampler can still be inefficient when its frozen proposal poorly covers target mass (Mengersen and Tweedie, 1996). Its acceptance and flow diagnostics therefore remain mandatory.

The two evaluation modes have different claims. Dynamic programming gives an exact target but requires the model. Fixed-tape sample-average inference needs only a generative simulator, but its finite- K target differs from the scientific target and may require many rollouts when β is large. It is therefore more accurate to call the overall approach simulator-based planning than model-free reinforcement learning. Bounded finite-horizon returns do permit a nonnegative Poisson estimator of $\exp(\beta J)$, and hence an exact pseudo-marginal construction; Appendix 8 gives the identity. Its relative variance can grow prohibitively at the large β values and return ranges needed here, so an implementation would require a separate variance-and-mixing gate rather than an unqualified claim of exact practicality.

The direct ELBO avoids exponentiating a noisy value estimate, but its score-function

gradient can have high variance and a factorized variational family cannot represent correlations between policy modes. Replicated SMC can retain a multimodal weighted population, yet it is vulnerable to particle degeneracy and must copy every memoized assignment and simulator replica during resampling.

All reported learned policies are stationary in physical state. This keeps the output class consistent but prevents any method from representing an action that depends explicitly on time-to-go. PPO uses Monte Carlo rather than generalized-advantage estimates and does not use value clipping. Discrete SAC fixes its entropy temperature but uses time-indexed twin critics and replay tuples for finite-horizon backups. Double Q-learning likewise uses time-indexed tables, then forms a stationary output by state–time-visitation-weighted projection. These are precise implemented variants, not claims of reproducing every enhancement of the algorithm families. GridWorld currently uses $\gamma = 1$, and the guard against unvalidated discounted runs must remain until the gradient and target semantics are tested.

Finally, GridWorld supports controlled correctness and scaling experiments but cannot establish broad generality. Blackjack is a low-cost nonspatial check. Tireworld would add structured stochastic failure at higher implementation cost. Factored concurrent-action and continuous-control domains require new policy representations and reference measures and are left to future work.

8 Conclusion

Expected-return policy inference is well defined only after the policy space, reference measure, inverse temperature, and execution semantics are explicit. The key mathematical choice is to exponentiate a policy’s expected return, not to average exponentiated realized returns. Guided replica-exchange Metropolis–Hastings provides an exact sampler when policy values are computed by dynamic programming and an exact sampler for a declared fixed-tape sample-average target under simulator access. Full-support guidance can improve proposals without changing either target because every local move is Hastings-corrected.

Bounded returns also yield a nonnegative unbiased Poisson weight for an exact simulator-only pseudo-marginal extension. Its explicit relative-variance formula shows why mathematical exactness alone is insufficient: at large inverse temperature it may require prohibitive rollout counts. It is therefore a gated extension, not a substitute for reporting the finite-tape approximation honestly.

The empirical claim remains deliberately narrower than the mathematical one. Exact tiny-domain agreement is established before scaling, and medium pilots must demonstrate mixing and acceptable cost before any confirmatory comparison. The final study will compare committed-policy control performance, target fidelity, and total compute against five precisely specified baselines. Until those gates pass, the contribution is a testable formulation and algorithmic protocol rather than a claim of empirical superiority.

Core Derivations

Gibbs variational identity. Let P_β be the target in Equation (2). For any $Q \ll \mu$,

$$\text{KL}(Q \| P_\beta) = \mathbb{E}_Q \left[\log \frac{dQ}{d\mu} - \beta J_H(\pi) + \log Z_\beta \right] \quad (19)$$

$$= \log Z_\beta - (\beta \mathbb{E}_Q[J_H(\pi)] - \text{KL}(Q \| \mu)). \quad (20)$$

Non-negativity yields Equation (4), with equality at $Q = P_\beta$.

Why a rollout is valid in the direct objective. For $\pi \sim q_\theta$ and $\tau \sim p(\cdot \mid \pi)$, iterated expectation gives

$$\mathbb{E}_{\pi, \tau}[G_H(\tau)] = \mathbb{E}_{\pi \sim q_\theta}[J_H(\pi)].$$

No exponential or logarithm is applied between the inner and outer expectations. Applying the score-function identity to Equation (16), and dropping terms whose expected score is zero, gives Equation (17).

Local detailed balance. At inverse temperature b , write

$$\tilde{P}_b(\pi) \propto \mu(\pi) \exp(b\tilde{J}(\pi)).$$

The Metropolis–Hastings ratio is

$$\frac{\tilde{P}_b(\pi')q(\pi \mid \pi')}{\tilde{P}_b(\pi)q(\pi' \mid \pi)} = \exp[b\{\tilde{J}(\pi') - \tilde{J}(\pi)\}] \frac{\mu(\pi') q_s(a \mid a')}{\mu(\pi) q_s(a' \mid a)}.$$

The uniform decision-state selection cancels, producing Equation (9). Full support makes every one-coordinate change possible, and a finite sequence of such changes connects any two tabular policies.

Replica-swap ratio. Before a swap, the factors for replicas i and j are proportional to $\mu(\pi_i) \exp(b_i \tilde{J}(\pi_i)) \mu(\pi_j) \exp(b_j \tilde{J}(\pi_j))$. After exchanging the two policies, the reference-measure factors cancel and the new-to-old density ratio is

$$\exp[(b_j - b_i)\{\tilde{J}(\pi_i) - \tilde{J}(\pi_j)\}],$$

which gives Equation (15). Alternating disjoint adjacent pairs changes scheduling efficiency but not this invariant product distribution.

Fixed-tape convergence. For every fixed policy, the strong law gives $\hat{J}_K(\pi) \rightarrow J(\pi)$ almost surely. Because the tabular policy space is finite, convergence holds simultaneously for all policies. Continuity of exponentiation and finite normalization then imply $P_{\beta, K}(\pi \mid \Xi_K) \rightarrow P_\beta(\pi)$ for every policy. This argument justifies a convergence study in K ; it does not make a finite tape bank exact for P_β .

A nonnegative unbiased policy weight from bounded rollouts. Suppose a simulator return satisfies $G_\pi \geq a$ almost surely, where the finite-horizon reward specification supplies the known constant a . For $\lambda > 0$, draw $N \sim \text{Poisson}(\lambda)$ and, conditional on N , independent returns $G_{\pi, 1}, \dots, G_{\pi, N}$. Define

$$\widehat{W}_\lambda(\pi) = \exp(\beta a) \prod_{i=1}^N \left(1 + \frac{\beta(G_{\pi, i} - a)}{\lambda}\right). \quad (21)$$

Every factor is nonnegative. Conditioning first on N , then using the probability generating function of a Poisson variable, gives

$$\mathbb{E}[\widehat{W}_\lambda(\pi)] = \exp(\beta a) \mathbb{E}_N \left[\left(1 + \frac{\beta(J(\pi) - a)}{\lambda}\right)^N \right] \quad (22)$$

$$= \exp(\beta a) \exp(\beta(J(\pi) - a)) = \exp(\beta J(\pi)). \quad (23)$$

Thus a pseudo-marginal Metropolis–Hastings chain on the policy and estimator randomness has the exact policy target as its marginal (Andrieu and Roberts, 2009; Jacob and Thiery,

2015). The identity is not automatically an efficient algorithm. Its expected rollout count is λ , and direct calculation gives

$$\frac{\mathbb{E}[\widehat{W}_\lambda(\pi)^2]}{\exp(2\beta J(\pi))} - 1 = \exp\left(\frac{\beta^2 \mathbb{E}[(G_\pi - a)^2]}{\lambda}\right) - 1. \quad (24)$$

Keeping this relative variance bounded can therefore require a rollout count that grows quadratically with β and with the distance from the return lower bound. This explains why the exact construction is retained as a diagnostic and future sampler rather than silently replacing the practical fixed-tape approximation.

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